BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) – 2024
SEMESTER – IV

Course Code: BCA49108
Course Name: Object Oriented Programming in JAVA Lab
Weekly Contact Hours: 4P
Credits: 2
Total Allotted Hours: 60P

Course Objective: The aim of this course is to provide an introduction to object oriented programming (OOP) using the Java programming language. Its main objective is to teach the basic concepts and techniques which form the object oriented programming paradigm.

Pre-requisite(s): Knowledge of programming language.

Course Outcomes (COs): After the completion of the course, students would be able to:

CO1: Write and analyze programs related to the concept of OOPs.
CO2: Implement classes and create objects with emphasis on data hiding.
CO3: Experiment and analyze polymorphism and inheritances in Java.
CO4: Test and assess exception handling and multithreading of Java.
CO5: Simulate applet programming and compare different Collections.

Module 1: Object Oriented Design **[8H]**

Basic Java programs with different data types, branching and iterative statements, compilation and interpretation of Java programs.

Module 2: Class & Object Properties **[8H]**

Classes and objects, implementation of access control modifiers in class

Module 3: Polymorphism and Inheritance **[20H]**

Programs with command line arguments, array and string, function overloading and overriding, constructor, constructor overloading, static data members and methods, inheritance by extending classes, inheritance with interfaces, packages

Module 4: Exception Handling & Multithreading **[12H]**

Thread using Thread class and Runnable interface, try, catch, throw and throws examples, user defined exception handling.

Module 5: Applet Programming and advanced topics **[12H]**

Array List, Vector, List, Map, HashMap, Set, HashSet and their detailed operations.

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Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K4, K5
2.	CO2	M2	K5, K6
3.	CO3	M3	K4
4.	CO4	M4	K4, K5
5.	CO5	M5	K4. K5, K6

Text Books:

1. Programming with Java: A Primer, E. Balagurusamy, TMH, 3rd Edition, 2006.